

3.4.1 Reactions of acids

Students should have knowledge and understanding of the following content.

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Outcome 1			
Acids react with some metals to produce salts and hydrogen. Hydrochloric acid produces chlorides and sulfuric acid produces sulfates. Students should be able to complete word equations for these reactions, given the names of the reactants.	Knowledge of reactions is limited to those of magnesium, zinc and iron with hydrochloric and sulfuric acids.	5.4.2.1	4.7.3.1
The test for hydrogen uses a burning splint held at the open end of a test tube of the gas. Hydrogen burns rapidly with a pop sound.	Suggested activity for TDA Investigate the amount of hydrogen produced when acids react with different metals.	5.8.2.1	4.7.3.1
Outcome 2			
Acids are neutralised by alkalis (eg sodium hydroxide) and bases (eg insoluble metal oxides) to produce salts and water.	Household chemicals may be used to illustrate these reactions.	5.4.2.2	4.7.3.2

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Acids are neutralised by carbonates to produce salts, water and carbon dioxide. Students should be able to complete word equations for these reactions, given the names of the reactants. Carbon dioxide turns limewater milky.	Suggested activity for TDA Investigate the reactions of acids with different carbonates.	5.4.2.2	4.7.3.1
Salt solutions can be crystallised to produce solid salts.		5.4.2.3	4.7.3.2